



BACKGROUND

Keywords: Automatic Speech Recognition (ASR), People Who Stutter (PWS), Word Error Rate (WER), Character Error Rate (CER)

ASR systems are widely used in technologies like voice assistants and translators. However, over 80 million PWS face significant challenges as ASR struggles to transcribe their speech accurately. This study explores: **How can fine-tuning ASR models improve recognition of disfluent speech?**

We evaluated ASR performance on stuttered speech with various stutter types in English and Mandarin, fine-tuning models to reduce bias i.e. inequalities in transcription accuracy between fluent and stuttered speech. By enhancing accuracy for stuttered speech, we aim to make ASR technology more inclusive and accessible.

DATA & METHODS

Datasets

To evaluate ASR models on stuttered speech, we utilized two datasets:

- LibriStutter [1]: 20 hours of English audio from 50 speakers, labeled with stutter types including sound repetition, word repetition, phrase repetition, and prolongation.
 - StammerTalk [2]: 50 hours of Mandarin audio from 72 speakers, labeled with stutter types such as word repetition, sound repetition, blocks, prolongation, and interjection.
- For training, the first 5000 audio samples from LibriStutter and the first 8000 samples from StammerTalk were used to fine-tune the models.

Stutter Types

- Word Repetition: Repeating entire words (e.g., "I-I-I want").
- Sound Repetition: Repeating single sounds (e.g., "b-b-b-ball").
- Phrase Repetition: Repeating full phrases (e.g., "I like I like I like").
- Blocks: Complete stoppage of speech, often with tension.
- Prolongation: Stretching sounds out (e.g., "ssssee").
- Interjection: Adding filler sounds or words (e.g., "um," "uh").

Model Training Pipeline

We fine-tuned OpenAI's Whisper-base ASR model [3] for both datasets:

- English Model: Learning rate = 1e-6, Dropout rate = 0.20
- Mandarin Model: Learning rate = 5e-6, Dropout rate = 0.20

Fine-tuning involved adjusting the model's parameters to improve its transcription accuracy for stuttered speech patterns. Fine-tuning refers to refining a pre-existing ASR model by training it further on specific datasets to better handle unique challenges like stuttering.

Evaluation Metrics

Model performance was measured using:

- Word Error Rate (WER) for English:
 - $WER = \frac{Insertions + Deletions + Substitutions}{Total Words in Ground Truth}$
 - Measures errors at the word level, suitable for English's word-based structure.
- Character Error Rate (CER) for Mandarin:
 - $CER = \frac{Insertions + Deletions + Substitutions}{Total Characters in Ground Truth}$
 - Measures errors at character level, suitable for Mandarin's monosyllabic structure.

This ensures evaluations are accurate for each language.

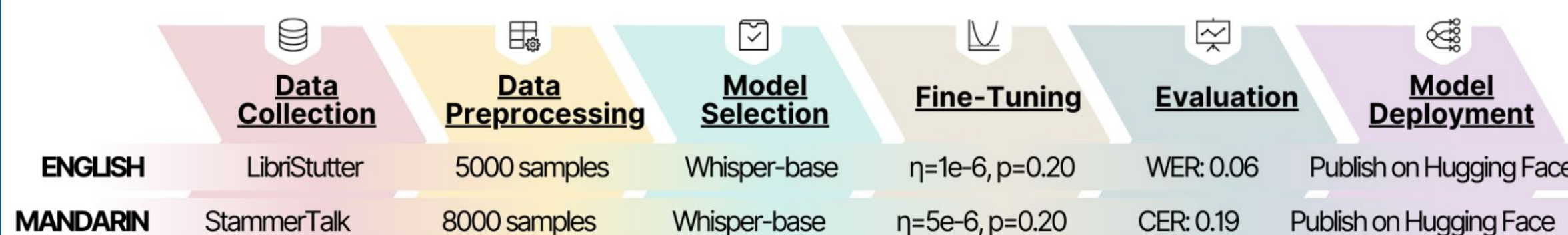
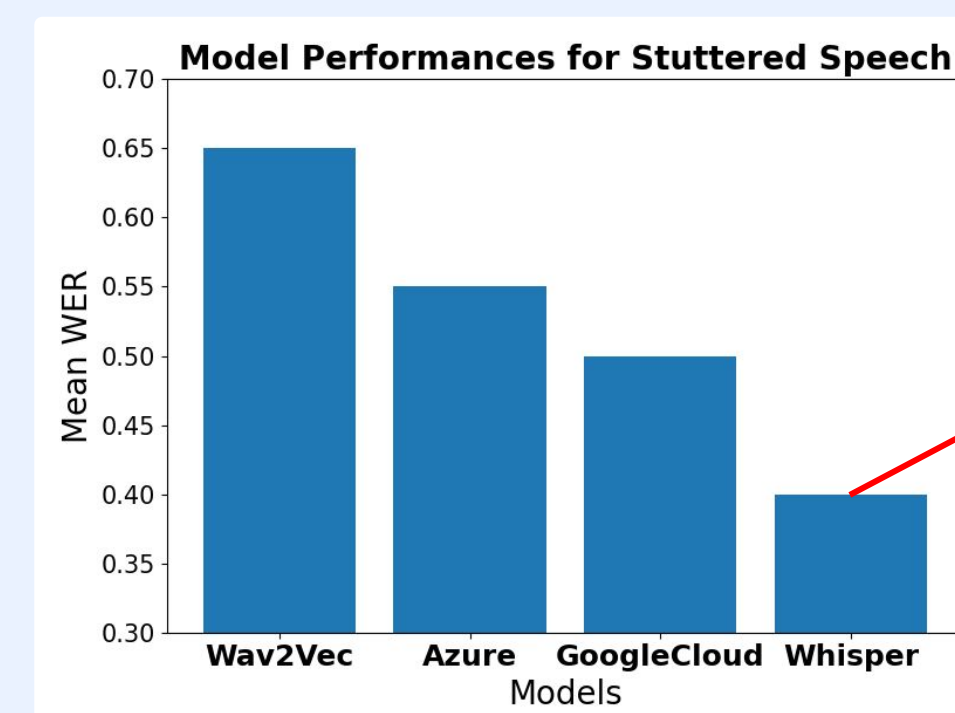


Figure 1. ASR fine-tuning workflow. Datasets were preprocessed at 16,000 Hz. Whisper-base was fine-tuned with learning rate (η) and dropout rate (p), evaluated using WER and CER, and published on Hugging Face.

RESULTS



Whisper achieves the lowest WER and outperforms all evaluated ASR models on stuttered speech, which made it the ideal baseline for fine-tuning in our study.

Figure 2. Mean WERs of 4 ASR models (Wav2Vec, Azure, Google Cloud Speech-to-Text, and Whisper) on stuttered speech are evaluated. Whisper has the lowest WER among all evaluated models, highlighting its robust performance as a baseline.

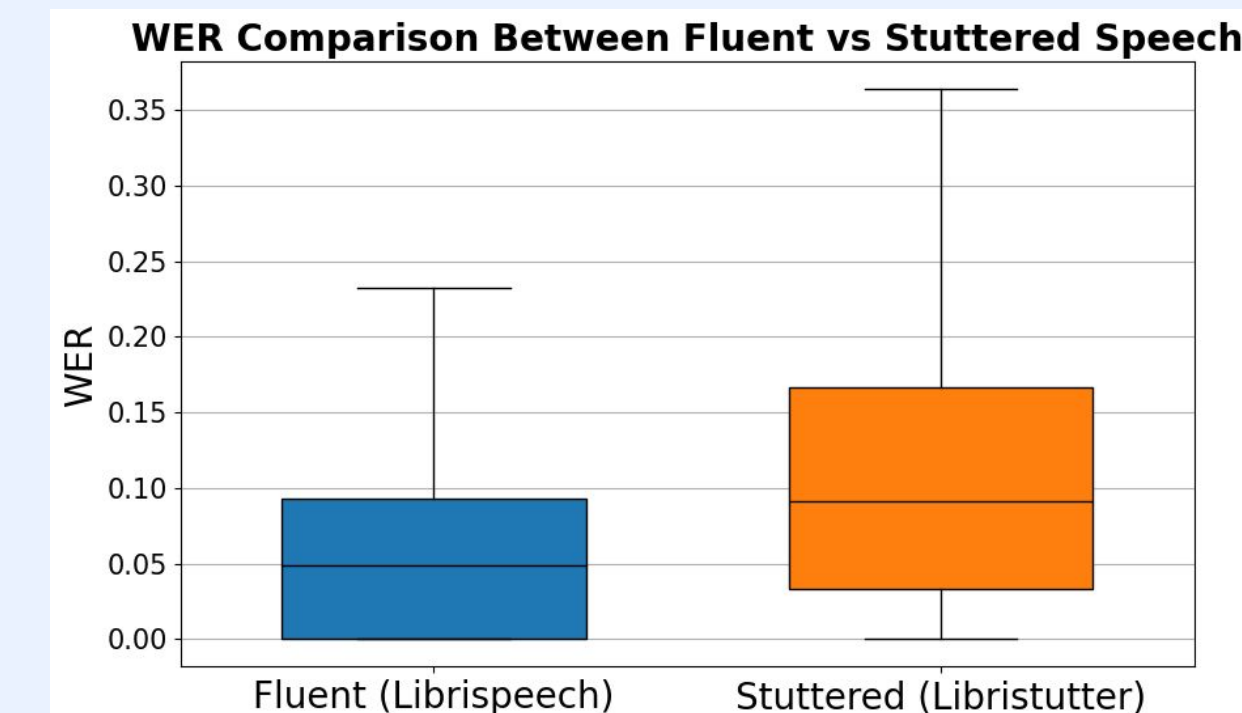


Figure 3. Shows the baseline Whisper's WER on fluent speech versus stuttered speech. The results show a significantly higher WER for stuttered speech, indicating model's struggle handling disfluent speech.

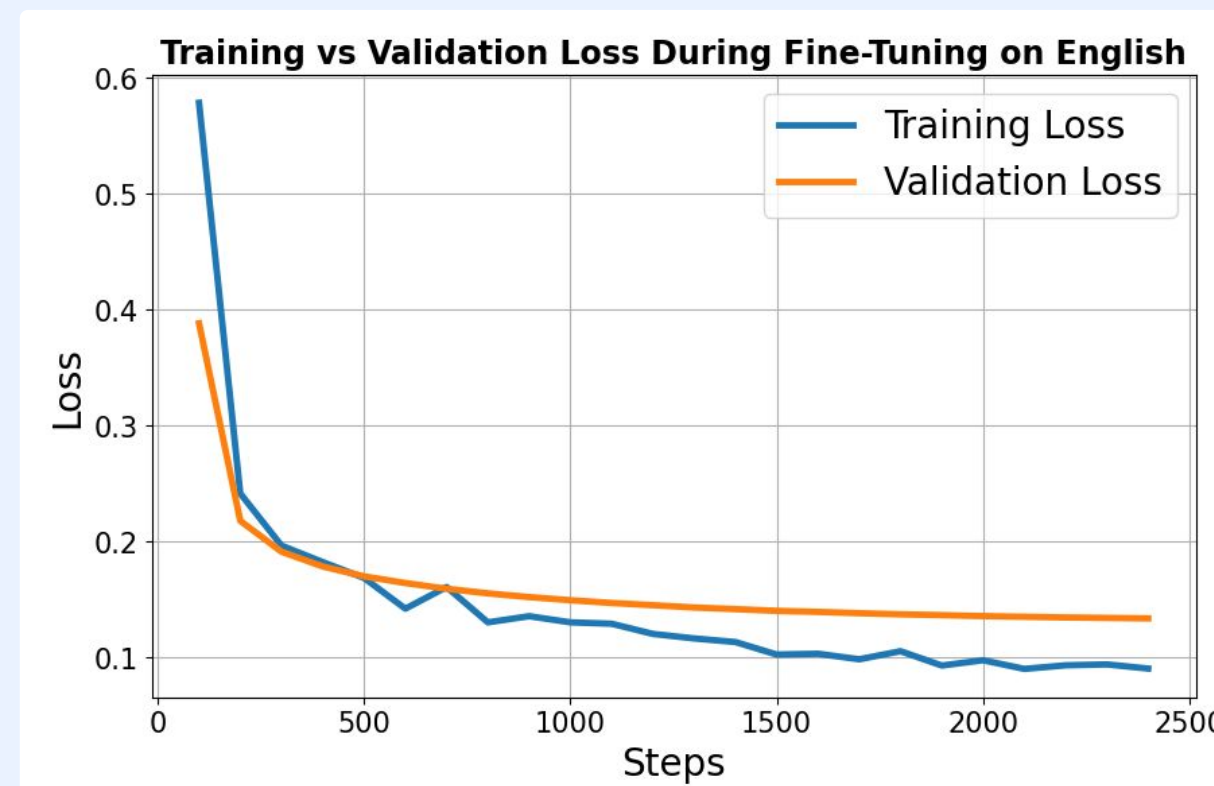


Figure 4. Training and validation loss curves during fine-tuning Whisper on English stuttering dataset. The steady decrease in losses indicates improved model performance without overfitting.

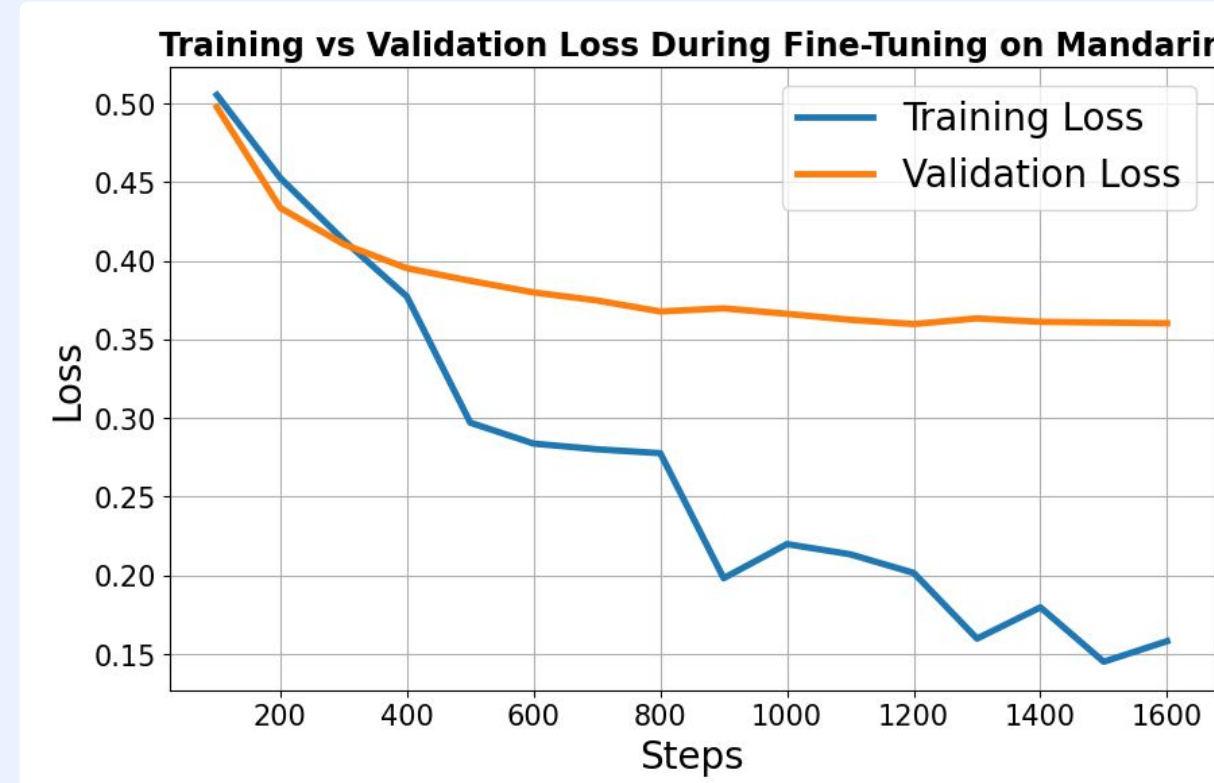


Figure 6. Training and validation loss curves during fine-tuning Whisper on Mandarin stuttering dataset.

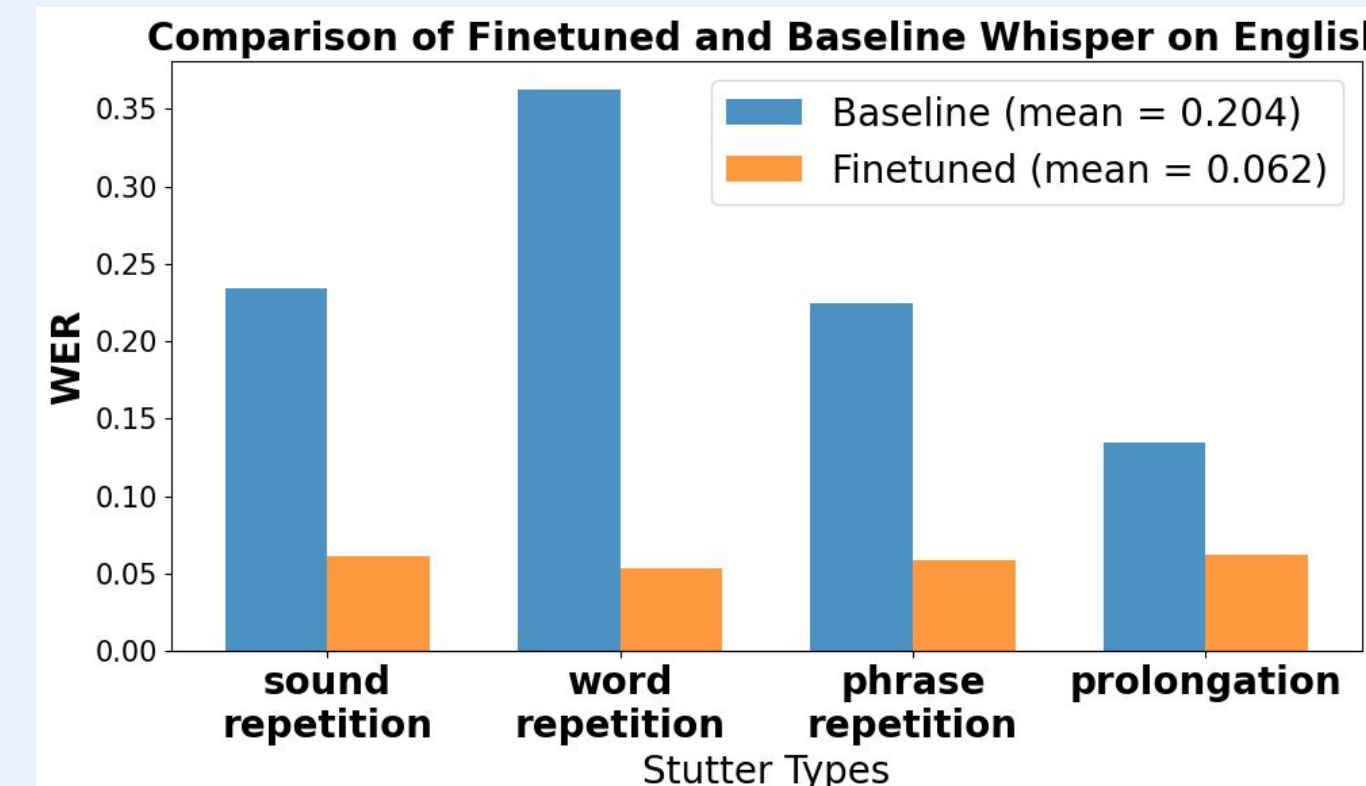


Figure 5. Compares the performance (WER) of the baseline and the fine-tuned Whisper model across various stutter types, including sound/word/phrase repetition and prolongation.

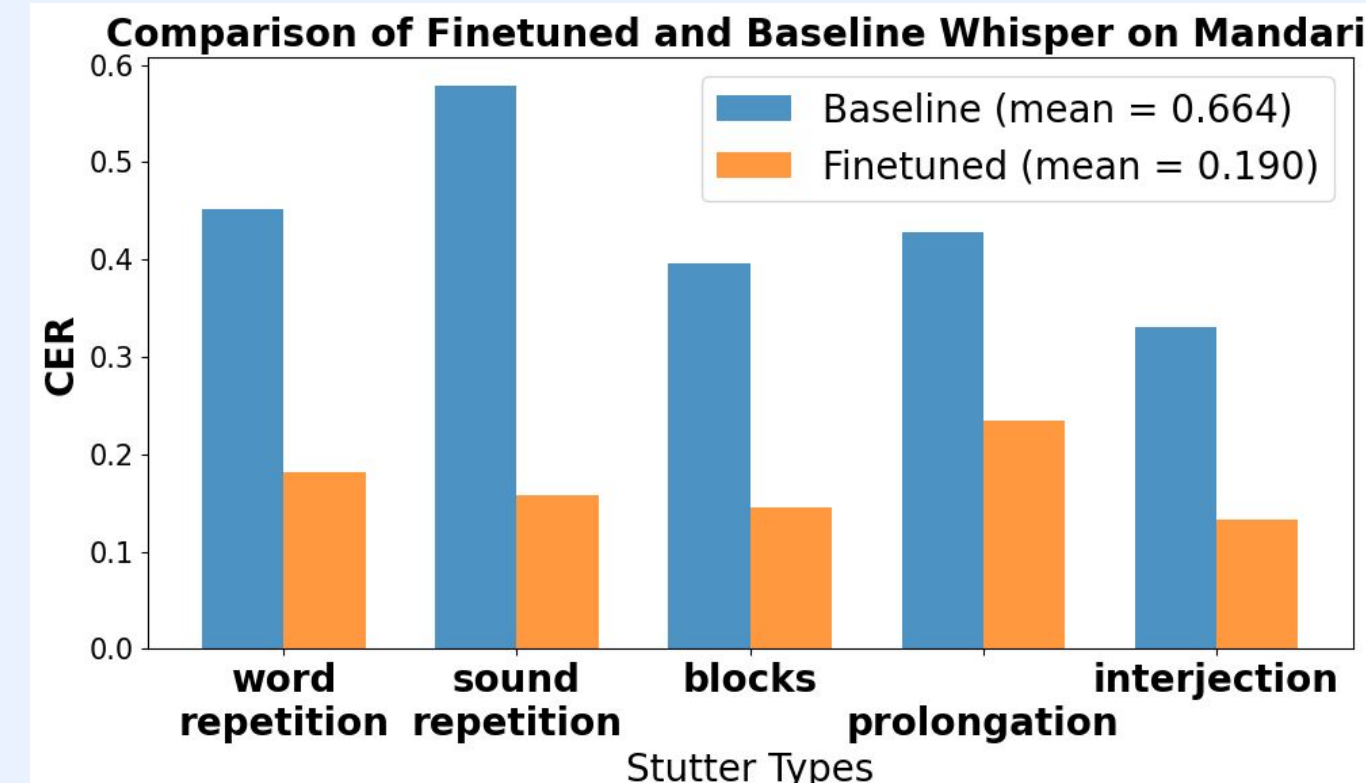


Figure 7. Compares baseline and fine-tuned Whisper model's performance (CER) across stutter types—word repetition, sound repetition, blocks, prolongation, and interjection.

ENGLISH MODEL

Fine-tuning reduces the mean WER from **20.4%** to **6.2%**, significantly improving the recognition of stuttered English speech. While the baseline model struggles most with word repetition, the fine-tuned model achieves consistently low WER across all stutter types, demonstrating its robustness and adaptability.

MANDARIN MODEL

Fine-tuning reduces the mean CER from **66.4%** to **19.0%**, significantly improving the recognition of stuttered Mandarin speech. While the base model struggles most with sound repetition, the fine-tuned model achieves consistently low CER across all stutter types, also demonstrating robustness and adaptability.

CONCLUSIONS

Validation was conducted using a subset of data from the LibriStutter dataset (English) and the StammerTalk dataset (Mandarin) for accurate comparisons between model transcriptions and ground truth. Our fine-tuned models achieved a **14.2% decrease in WER for English**, and a **47.4% decrease in CER for Mandarin** compared to the baseline model. These result highlight that our fine-tuning was extremely effective. The fine-tuned models are publicly available on Hugging Face [4][5] for anyone to use, enabling researchers, developers, and communities to build upon our work and further improve ASR systems for disfluent speech.

These results, achieved after six months of focused research and application, show a glaring oversight in ASR systems developed by large corporations including OpenAI's Whisper. Despite their immense resources, they failed to adequately account for diverse speech patterns. **Our model sets a precedent for making AI systems more inclusive from the ground up.** This work could serve as a catalyst for redefining industry standards, ensuring future ASR systems are built with a deeper understanding of diverse speech characteristics, ultimately creating more equitable AI solutions for all users, including the 80 million PWS.

KEY FINDINGS

- Sound/word repetitions were the worst performing stutter types for both English and Mandarin but showed the greatest improvement after fine-tuning, with error rates comparable to other stutter types. This demonstrates that the fine-tuned models effectively address the repetitive nature of stuttered speech.
- For the English fine-tuned model, the WER decreased to around 5% across all stutter types, showing that fine-tuning effectively bridges the performance gap for stuttered English speech.
- The Mandarin fine-tuned model showed more variability in error rates across different stutter types, with a mean CER of 19%. Prolongation had the highest error rate, likely due to Mandarin's tonal nature, where tones play a key role in distinguishing meaning. Elongated sounds can disrupt tonal patterns, introducing ambiguity and posing challenges for accurate transcription.
- The study highlights the importance of linguistic factors, such as tonal variations in Mandarin and repetitive patterns in stuttering, in ASR performance.

FUTURE WORK

- Expanding Inclusivity:** Fine-tune speech recognition models for diverse languages, dialects, accents, and stutter types.
- Addressing Dataset Scarcity:** Collaborate with institutions and communities to create publicly available stuttered speech datasets covering diverse patterns.
- Enabling Public Access:** Provide fine-tuned models to the public through open-source platforms for practical use.

CITATIONS

- [1] Kourkounakis, T. (2021). *LibriStutter* (Version V1) [Dataset]. Borealis. <https://doi.org/10.5683/SP3/NKVOGQ>
- [2] StammerTalk, Almpower, AIShell, Northwestern Polytechnical University, & Wenet Community. (2023). *StammerTalk-speech-70 Dataset* [Dataset]. Licensed under CC BY-NC 4.0.
- [3] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). *Robust speech recognition via large-scale weak supervision*. arXiv. <https://arxiv.org/abs/2212.04356>
- [4] Fine-tuned Whisper English model. Available at: <https://huggingface.co/dongim04/whisper-base-en>
- [5] Fine-tuned Whisper Mandarin model. Available at: <https://huggingface.co/dongim04/whisper-base-zh>

Ground Truth	Whisper Base Transcription WER: 59%	Fine-Tuned Transcription WER: 0%
said ronicky doone (Word Repetition) bill look me in the eye and tell me man to man that you're a liar he added can you ever be (Phrase Repetition) happy without her	said ronnyky dune dune dune bill look me in the eye and tell me man to man the cheerleyer he added can you ever be can you ever be happy without her	said ronicky doone bill look me in the eye and tell me man to man that you're a liar he added can you ever be happy without her

Table 8. Example comparing baseline and fine-tuned model transcriptions with ground truth. The base model struggles with repeated words and phrases, while the fine-tuned model accurately removes repetitions, showing its effectiveness in handling disfluencies.

Instructions

▼ Steps and Specifications to Create Your ePoster

These instructions are written for PowerPoint, but are applicable to any other software you may wish to use. If you use PowerPoint, please use PowerPoint 2007 or newer.

Creating an ePoster Upload

1. Full Instructions - [Download](#)
2. Number of pages (slides): one (1)
3. Max file size accepted: 55120KB (1024KB = 1MB)
4. Set the dimensions of your poster to 43.6" (110.7 cm) width x 24.5" (62.2 cm) height in landscape orientation, or start with

[this template](#) .

Note: We highly encourage students to use this template to reduce any chance of blurriness. Do not feel restricted to the colors shown in the template. The intent of the template is merely to ensure the dimensions are appropriate for the display screens that will be used at the meeting.

To set your slide dimensions:

- Select the "Design" tab in PowerPoint
 - Click "Slide Size" and select "Custom Slide Size"
 - Set the width and height as outlined above
5. Fill in the abstract title, author names, and institutional affiliations of the authors at the top of the poster.
 6. Fill in content as appropriate.
 7. **Before submitting, save your poster as a PDF file. All recent versions of PowerPoint and most other software applications allow you to save your poster as a PDF file from the "File > Save as" menu.**

PLEASE NOTE: Using the "File > Print > as .PDF" option may make your poster blurry. If you created the pdf this way and are seeing blurriness, please try using the "File > Save as" option.

▼ Requirements & Guidelines

Following these guidelines will ensure that your ePoster is readable and effective.

- ✓ Make sure your text and background have a **large contrast** (dark lettering on a light background or the reverse).
- ✓ A **minimum 22-28 point size (or bigger)** for body text is used to ensure optimal legibility from the usual distance of 3-5 ft.
- A little larger text (e.g.: up to 32 pt size) might also be a good idea, as it will provide comfortable reading from an even larger distance, up to 8 ft.

✓ **For embedded images we recommend** using .jpg or .png file formats in a resolution of 72 or 96 dpi.

✓ **Do not use animated effects, "animations," or videos.**

✓ **Save space for controls:** When the full-size ePoster is displayed at the meeting, there will be some controls covering a small area in the lower right-hand corner, and some additional information will be displayed in a small area in the upper right-hand corner. We recommend leaving those areas blank (as shown on the template).

BACKGROUND

Keywords:

Automatic Speech Recognition (ASR), People Who Stutter (PWS), Word Error Rate (WER), Character Error Rate (CER)

ASR is widely used in tools like voice assistants and translators but often struggle to transcribe stuttered speech, impacting 80 million PWS. Our research question is: **How can fine-tuning ASR models improve recognition of disfluent speech?**

We evaluated models on English and Mandarin stutters and fine-tuned them to reduce **bias¹** to make ASR technology more inclusive and accessible.

¹ Inequalities in transcription accuracy between fluent and stuttered speech

DATA & METHODS

Datasets

- LibriStutter [1]: 20 hours of **English** stuttered speech audio.
- StammerTalk [2]: 50 hours of **Mandarin** stuttered speech audio.

Stutter Types

1. **Word Repetition**: Repeating entire words (e.g., "I-I-I want").
2. **Sound Repetition**: Repeating single sounds (e.g., "b-b-b-ball").
3. **Phrase Repetition**: Repeating full phrases (e.g., "I like I like I like").
4. **Blocks**: Complete stoppage of speech, often with tension.
5. **Prolongation**: Stretching sounds out (e.g., "ssssee").
6. **Interjection**: Adding filler sounds or words (e.g., "um," "uh").

Model Training Pipeline

We fine-tuned **OpenAI's Whisper-base** ASR model [3] for each of the English and Mandarin dataset. Fine-tuning improves model performance by further training it on specific datasets to better handle stuttering.

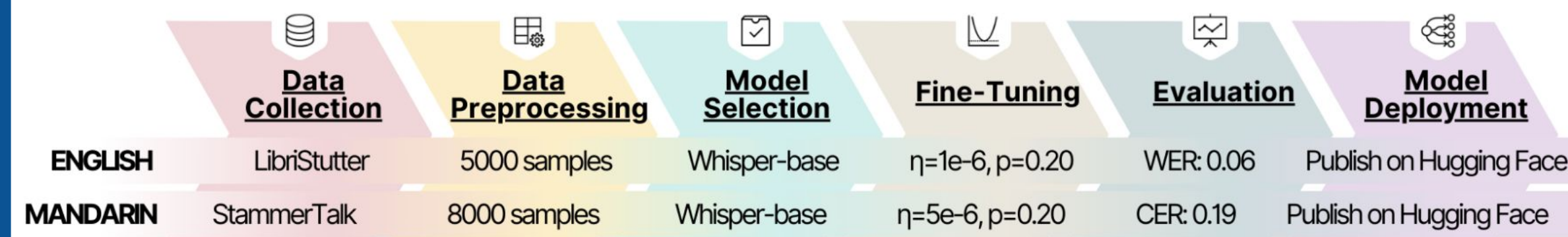


Figure 1. ASR fine-tuning workflow. Datasets were preprocessed, and Whisper-base was fine-tuned with learning rate (η) and dropout rate (p), evaluated, and published.

Evaluation Metrics

- **WER**(Word Error Rate) → Used for **English's** word-based structure
- **CER**(Character Error Rate) → Used for **Mandarin's** monosyllabic structure

$$WER(CER) = \frac{\text{Insertions} + \text{Deletions} + \text{Substitutions}}{\text{Total Words(Characters) in Ground Truth}}$$

RESULTS

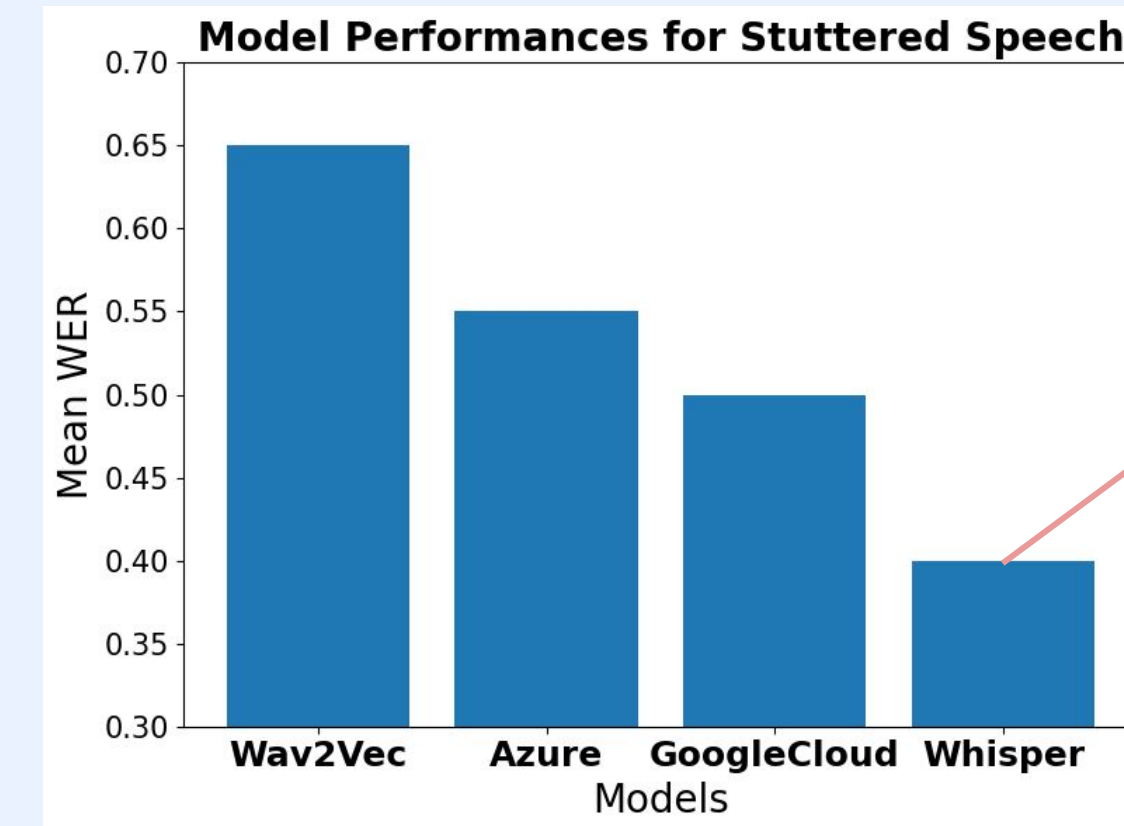


Figure 2. Mean WERs of ASR models on stuttered speech.

Whisper achieves the lowest WER and outperforms all evaluated ASR models on stuttered speech, which made it the ideal baseline for fine-tuning in our study.

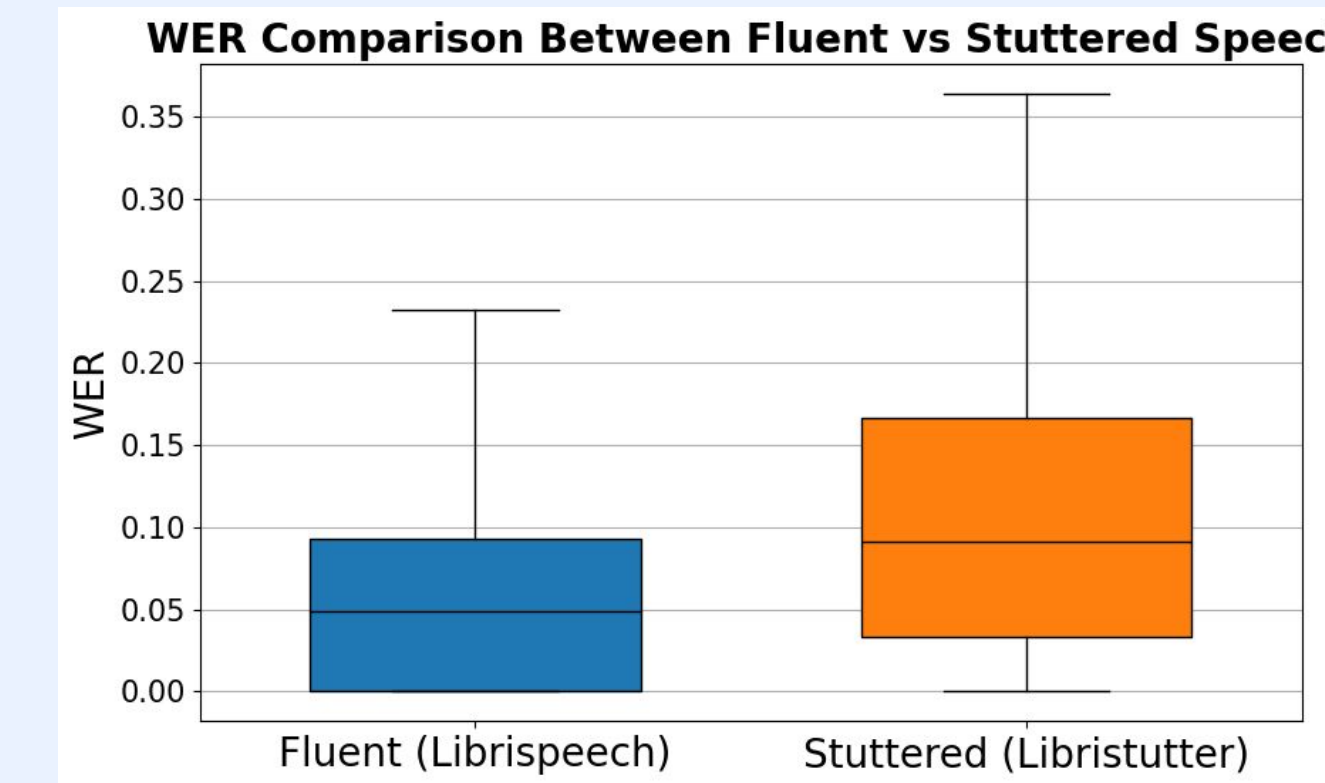


Figure 3. Baseline Whisper shows higher WER for stuttered speech, indicating difficulty with disfluency.

ENGLISH MODEL: WER 20.4% → 6.2% (-14.2%)

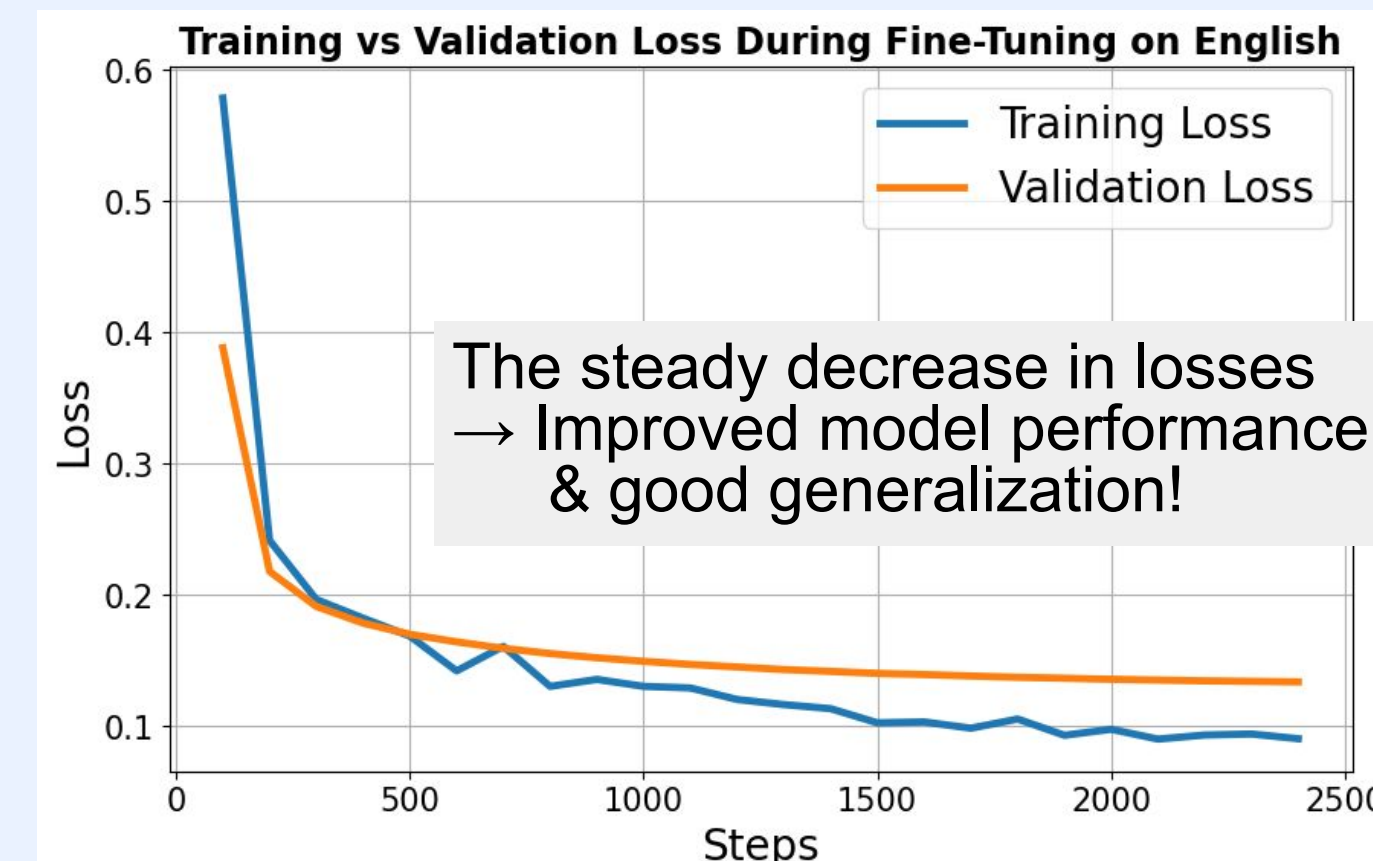


Figure 4. Train/validation loss for English fine-tuning.

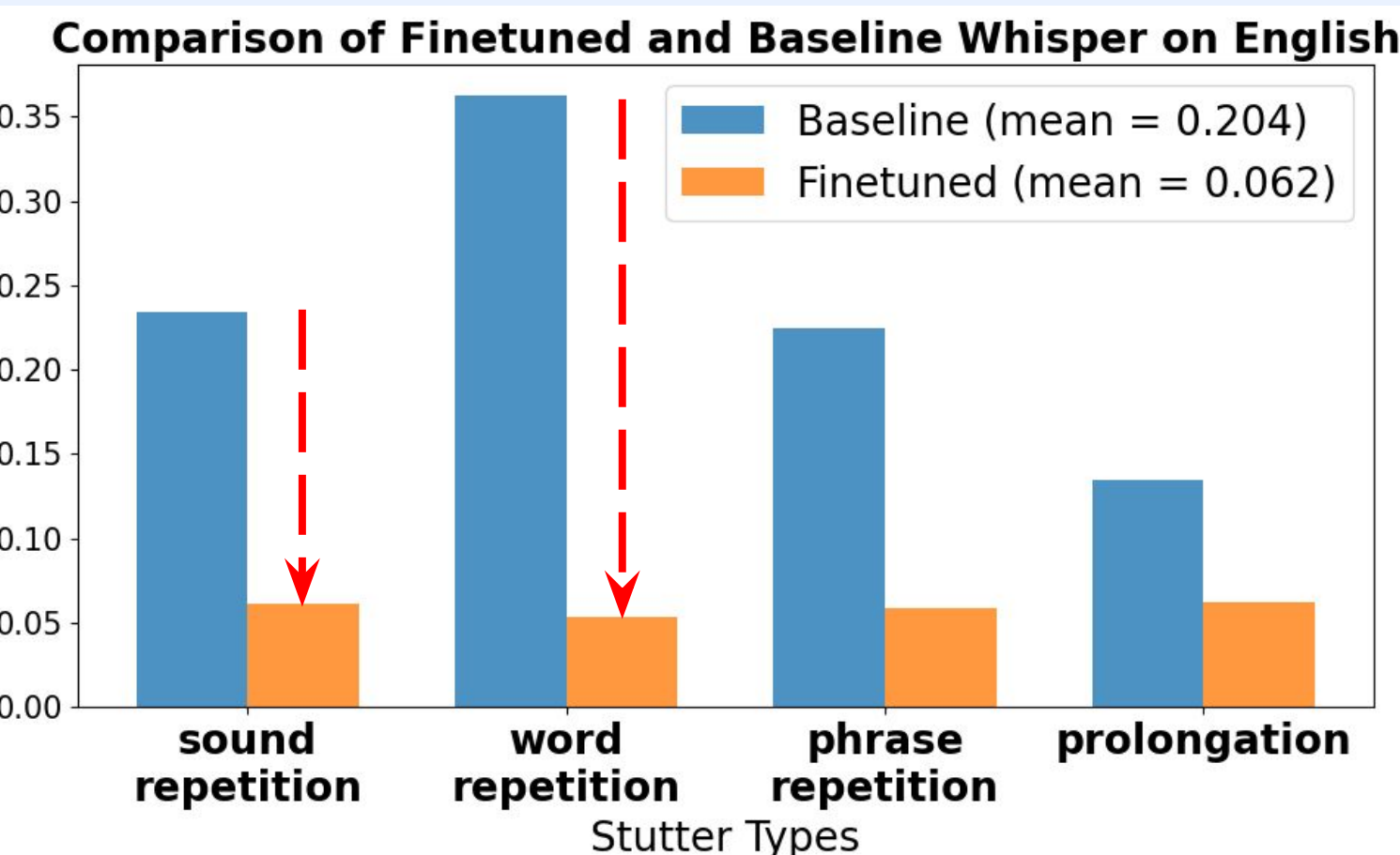


Figure 5. Compares baseline and fine-tuned English model's WER across stutter types.

MANDARIN MODEL: CER 66.4% → 19.0% (-47.4%)

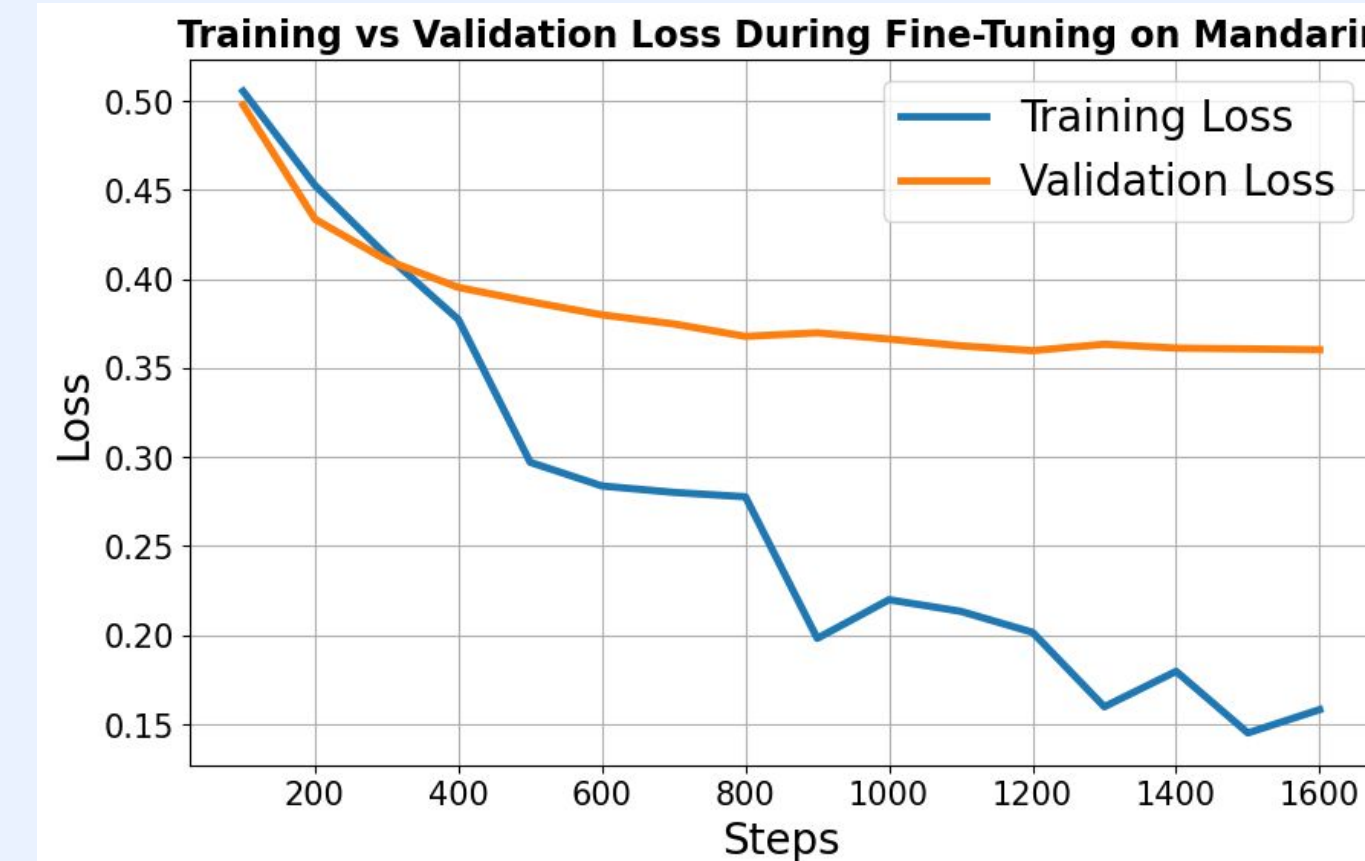


Figure 6. Train/validation loss for Mandarin fine-tuning.

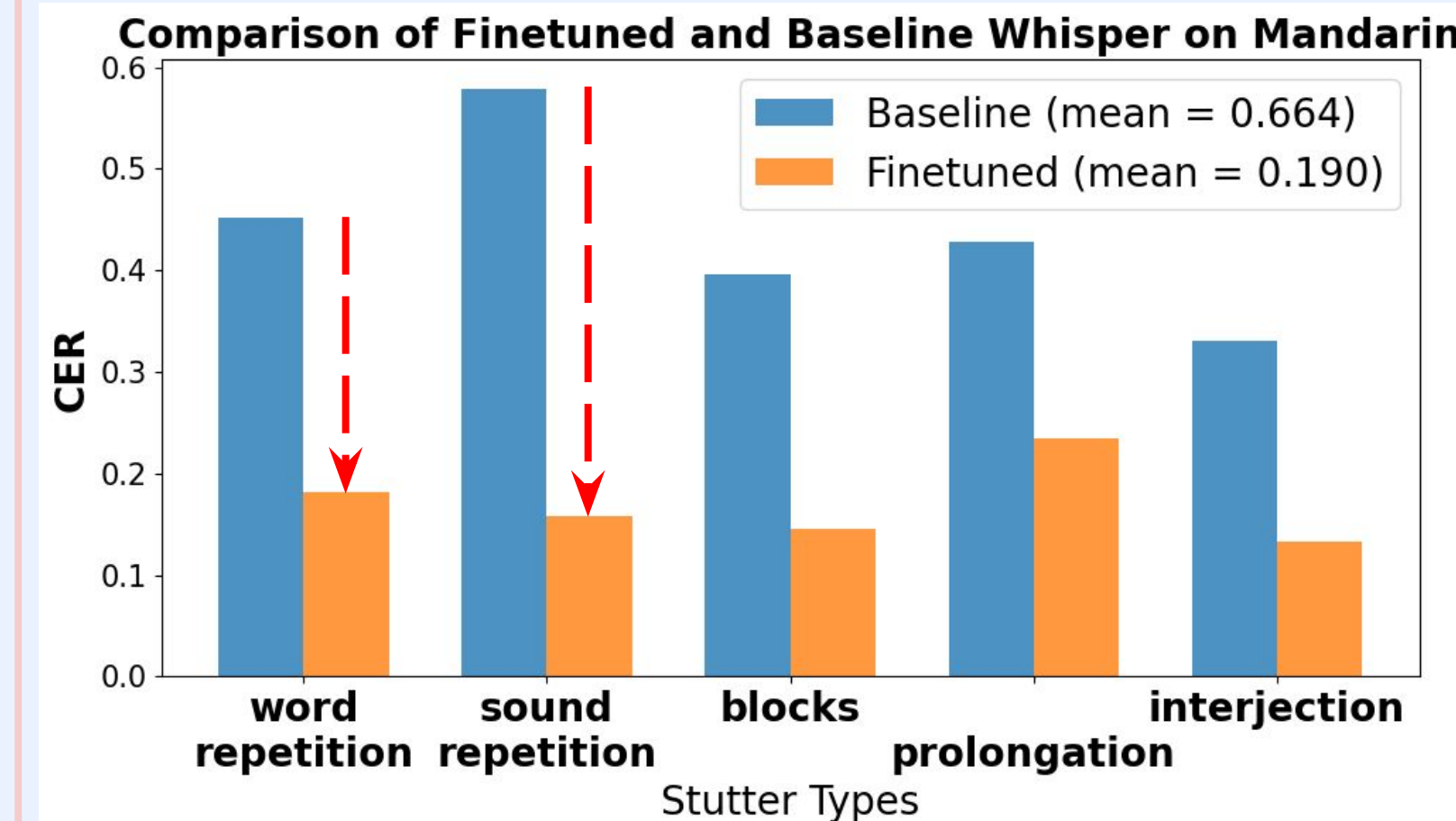


Figure 7. Compares baseline and fine-tuned Mandarin model's CER across stutter types.

Ground Truth	Whisper Base Transcription WER 59%	Fine-Tuned Transcription WER: 0%
... doone (Word Repetition) ... to man that you're a liar he added can you ever be (Phrase Repetition) happy ...	... dune dune dune ... the cheerleyer he added can you ever be can you ever be can you ever be happy ...	... doone ... to man that you're a liar he added can you ever be happy ...

Table 1. Example transcriptions from the baseline and fine-tuned models. The base model struggles with repetitions, while the fine-tuned model accurately removes repetitions, showing its effectiveness in handling disfluencies.

CONCLUSIONS

- **14.2% WER reduction** for the **English fine-tuned model** and **47.4% CER reduction** for the **Mandarin model**, proving their robustness and adaptability for stuttered speech.
* Fine-tuned models are available on Hugging Face [4][5].
- While even large corporations like OpenAI have not fully addressed diverse speech patterns in ASR systems, our work sets the foundation for **inclusive AI that supports marginalized communities**.
- This research encourages ASR developers to prioritize inclusivity and build equitable AI for all, including 80M PWS.

KEY FINDINGS

- Fine-tuning significantly improved **sound/word repetitions** → Our models successfully address the **repetitive nature of stuttered speech**
 - English Model: WER dropped to ~5% across all stutter types → Model performs stable on stuttered English
 - Mandarin Model: **Prolongation** had the highest CER → **Mandarin is tonal language**. Elongated sounds could disrupt tonal patterns, causing inaccurate transcription.
- **Linguistic factors** significantly impact ASR performance!

FUTURE WORK

- Fine-tune ASR models on diverse languages, dialects, and accents to expand inclusivity.
- Create publicly available stuttered speech datasets by collaborating with communities to address data scarcity.

CITATIONS

- [1] Kourkounakis, T. (2021). *LibriStutter* (Version V1) [Dataset]. Borealis. <https://doi.org/10.5683/SP3/NKVOGQ>
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- [3] Radford, A., Kim, J. W., Xu, T., Brockman, G., McLeavey, C., & Sutskever, I. (2022). *Robust speech recognition via large-scale weak supervision*. arXiv. <https://arxiv.org/abs/2212.04356>
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- [5] Fine-tuned Whisper Mandarin model. Available at: <https://huggingface.co/dongim04/whisper-base-zh>